



Solving and Analyzing Quadratic Inequalities

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Learning Objectives

- Solve quadratic inequalities by factoring and sign analysis
- Use a number line / test points to determine solution intervals
- Recognize special cases (perfect squares, no solution, all reals)
- Express solutions in interval notation

Solve each quadratic inequality completely, showing factoring and sign analysis, then write the solution in interval notation.

1. Solve and graph. Write the solution in interval notation:

$$x^2 - 5x + 6 < 0$$

Answer: _____

2. Solve:

$$x^2 - x - 12 \geq 0$$

Answer: _____

3. Solve:

$$x^2 - 9 < 0$$

Answer: _____

4. Solve:

$$x^2 + 2x - 8 > 0$$

Answer: _____

5. Solve:

$$x^2 - 4x \leq 0$$

Answer: _____

6. Solve:

$$2x^2 - 3x - 2 \leq 0$$

Answer: _____

7. Solve:

$$x^2 - 2x - 3 \geq 0$$

Answer: _____



8. Solve. Watch for a special case:

$$x^2 - 6x + 9 > 0$$

Answer: _____

9. Solve. Watch for a special case:

$$x^2 + 4 \leq 0$$

Answer: _____

10. Solve:

$$x^2 + 5x + 6 > 0$$

Answer: _____





Method: move all terms to one side, factor, find the boundary roots, then test the sign of each interval. Remember perfect-square and no-solution special cases.

Solutions

1. Solve and graph. Write the solution in interval notation:

$$x^2 - 5x + 6 < 0$$

→ Factor: $(x-2)(x-3) < 0$.

→ Roots 2 and 3; product negative between roots.

→ Solution: $2 < x < 3$.

Answer: $2 < x < 3 \Rightarrow (2, 3)$

2. Solve:

$$x^2 - x - 12 \geq 0$$

→ Factor: $(x-4)(x+3) \geq 0$.

→ Roots -3 and 4; product non-negative outside the roots.

→ Solution: $x \leq -3$ or $x \geq 4$.

Answer: $x \leq -3$ or $x \geq 4$

3. Solve:

$$x^2 - 9 < 0$$

→ Factor: $(x-3)(x+3) < 0$.

→ Negative between roots -3 and 3.

Answer: $-3 < x < 3 \Rightarrow (-3, 3)$

4. Solve:

$$x^2 + 2x - 8 > 0$$

→ Factor: $(x+4)(x-2) > 0$.

→ Positive outside the roots -4 and 2.

Answer: $x < -4$ or $x > 2$

5. Solve:

$$x^2 - 4x \leq 0$$

→ Factor: $x(x-4) \leq 0$.

→ Roots 0 and 4; non-positive between them.

Answer: $0 \leq x \leq 4 \Rightarrow [0, 4]$



6. Solve:

$$2x^2 - 3x - 2 \leq 0$$

→ Factor: $(2x+1)(x-2) \leq 0$.

→ Roots $-1/2$ and 2 ; non-positive between them.

Answer: $-\frac{1}{2} \leq x \leq 2 \Rightarrow \left[-\frac{1}{2}, 2\right]$

7. Solve:

$$x^2 - 2x - 3 \geq 0$$

→ Factor: $(x-3)(x+1) \geq 0$.

→ Non-negative outside the roots -1 and 3 .

Answer: $x \leq -1$ or $x \geq 3$

8. Solve. Watch for a special case:

$$x^2 - 6x + 9 > 0$$

→ Factor: $(x-3)^2 > 0$, a perfect square.

→ A square is > 0 everywhere except where it equals 0 .

→ Solution: all reals except $x = 3$.

Answer: $x \neq 3 \Rightarrow (-\infty, 3) \cup (3, \infty)$

9. Solve. Watch for a special case:

$$x^2 + 4 \leq 0$$

→ x squared is always ≥ 0 , so x squared $+ 4 \geq 4 > 0$ for all x .

→ The inequality is never satisfied.

Answer: No solution $\Rightarrow \emptyset$

10. Solve:

$$x^2 + 5x + 6 > 0$$

→ Factor: $(x+2)(x+3) > 0$.

→ Positive outside the roots -3 and -2 .

Answer: $x < -3$ or $x > -2$

